Life Cycle of a Star

Exam code: 0625 & 0972

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Life cycle of low mass stars

Extended tier only

- A low-mass star is one with a mass of up to about eight times that of the Sun
- After the main sequence, a low-mass star finishes its life cycle in the following evolutionary stages:

red giant → planetary nebula → white dwarf

Red giant

- After several **billion** years, the hydrogen fuel used for nuclear reactions begins to run out
- Once this happens, the rate of fusion decreases, which causes the core to shrink and heat up
  - As the energy produced by fusion decreases, the inward force due to gravity becomes **greater** than the outward force due to the thermal pressure
- Eventually, the star becomes a **red giant** when the core becomes hot enough for **helium** to fuse into **carbon**
- The energy released by re-ignited fusion reactions causes the outer layers of the star to **expand** and **cool**

Planetary nebula

- Once the helium in the core runs out, fusion reactions cannot continue

Help me understand

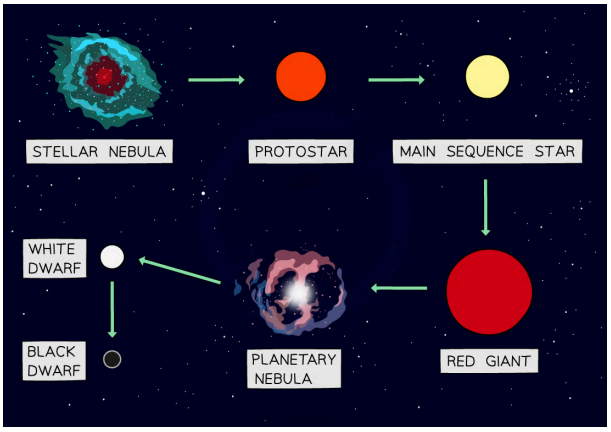
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

nebula

White dwarf

- The collapsed core of the red giant is called a **white dwarf**
- The white dwarf **cools** down over time and as a result, the amount of **energy** it emits **decreases**

The life cycle of a low-mass star



The life cycle of a star that is similar to our Sun

Examiner Tips and Tricks

A low mass star is any star that will eventually become a **white dwarf**. You may see different sources giving different ranges of masses for stars within this category, or terms such as low mass stars (up to 2 solar masses) or intermediate mass stars (between 2 and 8 solar masses). Note that you do not need to know these numbers or categories, only that all these stars will follow the **same** evolutionary stages.

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Did this video help you? Yes No

Extended tier only

- A high-mass star is one with a mass of more than about eight times that of the Sun
- After the main sequence, a high-mass star finishes its life cycle in the following evolutionary stages:

red supergiant → supernova → neutron star (or black hole)

Red supergiant

- After several **million** years, the hydrogen in the core begins to run out
- Similar to a low-mass star, the rate of fusion decreases and the core shrinks and heats up
- The star becomes a **red supergiant** when the core becomes hot enough for helium fusion to start
- This causes the outer layers of the star to **expand** and **cool**
- In the core of the star, **helium** fuses into **carbon**
  - This is followed by further fusion reactions in which successively heavier elements, such as nitrogen and oxygen, are formed
- During this stage, the core collapses and expands repeatedly as fusion reactions start and stop

Supernova

- Eventually, fusion reactions inside the red supergiant cannot continue once **iron** is formed
- The core of the star will **collapse rapidly** and initiate a **gigantic explosion** called a **supernova**
- At the centre of this explosion, a **dense** body called a **neutron star** will form
- The **outer layers** of the star are **ejected** into space forming **new clouds** of dust and gas (**nebula**)
- The nebula from a supernova may form new **stars** with orbiting **planets**
  - The heaviest elements (elements heavier than iron) are formed during a supernova and are ejected into space
  - These nebulae may form new planetary systems

Neutron star (or black hole)

- In the case of the **most massive stars**, the neutron star that forms at the centre will continue to **collapse** under the force of **gravity** until it forms a **black hole**
- A black hole is an **extremely dense** point in space that not even **light** can escape from

The life cycle of a high-mass star

The life cycle of a star much larger than our Sun

Examiner Tips and Tricks

A high mass star is a one that will **not** eventually become a white dwarf. Make sure you understand that most high mass stars become neutron stars and only the highest mass stars become black holes.

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